A DISCRETE-SINE-TRANSFORM BASED PRECONDITIONER FOR FRACTIONAL DERIVATIVE EQUATIONS ON ADAPTIVE MESHES *

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Abstract. We present a preconditioner for solving fractional differential equations (FDEs) on an adaptive mesh. Adaptive refinement of the problem domain results in a stiffness matrix with Toeplitz blocks along the main diagonal, while the FDE yields a dense stiffness matrix where off-diagonal blocks are stored as low-rank approximations. Our adaptive preconditioner utilizes ideas from the circulant preconditioner of Chan and Strang [SIAM Journal on Scientific Computing, 1989] and the discrete-sine-transform based preconditioner of Bini and Di Benedetto [Association for Computing Machinery, 1990]. We take advantage of the Toeplitz blocks on the diagonal and accounts for the low-rank nature of the off-diagonal blocks. We demonstrate the adaptive preconditioner's effectiveness at accelerating convergence for our systems and emphasize its efficient application. This work presents theoretical results about the spectral clustering of the preconditioned system. In order to prove these results, special consideration is taken on how the low-rank blocks perturb the eigenvalues of the Toeplitz block-diagonal system. Numerical tests are used to inspect any assumptions and validate the results.

Key words. Preconditioner, Fractional Derivative Equations, Adaptive Refinement, Finite-element Method, Toeplitz, Circulant, DST

MSC codes. 65M12, 65M50

1. Introduction.

In this paper we consider fractional differential equations (FDEs) of the form

$$D_x^\alpha u(x) = f(x),\ x \in (a,b), \tag{1.1}$$

where D_x^α is the Riesz fractional derivative of order $1 < \alpha < 2$ and $f \in L^2([a,b])$. Fractional equations find applications in many fields such as fluid mechanics, electromagnetic fluids, image processing, and material science (see for example [15, 9]). As with partial differential equations, the finite-element method (FEM) can be used to discretize FDEs into linear systems. This approach poses two main challenges. First, the stiffness matrix resulting from FEM discretization is dense due to the fractional order and non-locality of the operator. Second, by definition, fractional derivatives involve weak integration of singular kernels [16]. Thus solutions have singularities near the boundaries, making Galerkin finite-element methods sub-optimal [13, 18].

The first challenge can be overcome when using FEM on a uniform mesh, which imparts structure, producing a Toeplitz or Toeplitz-like stiffness matrix. This structure can be leveraged for efficient storage and linear solves. However, it is exactly this uniform mesh that exacerbates boundary singularities. The mesh must be made very fine to achieve uniform error below a given threshold in intervals near the boundary, causing the matrix size to grow. An optimal mesh would have the L^∞ error equally distributed on each interval, but a uniform mesh does not, in general, satisfy this condition. Alternatively, to achieve desired accuracy across the whole domain, we use an adaptive mesh and adaptive finite-element method (AFEM) [14, 19]. The usual AFEM algorithm iterates through the process

$$\texttt{SOLVE} \rightarrow \texttt{ESTIMATE} \rightarrow \texttt{MARK} \rightarrow \texttt{REFINE}. \tag{1.2}$$

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Here, the computational cost is dominated by the `SOLVE` step. The stiffness matrix, A, produced by AFEM is SPD, but also dense, ill-conditioned, and has lost its general Toeplitz-like structure.

Various techniques have been developed to address these complications on a uniform mesh. In [11], the authors use a reconstruction technique that isolates the singularity to improve finite-element method (FEM) convergence. Alternatively, in [18], a Toeplitz preconditioner for fractional diffusion equations reduces iteration cost to $\mathcal{O}(m \log m)$, where m is the size of A, using the preconditioned conjugate gradient squared method.

To our knowledge, the only work done on an adaptive mesh is from Zhao et al. [19]. Their technique uses AFEM to solve FDEs using hierarchical matrices and geometric multigrid (GMG) in $\mathcal{O}(kN \log N)$ where k is a parameter for low-rank approximations and N is the degrees of freedom at the finest level. Numerical tests show this technique attains optimal convergence order, but there are no proofs guaranteeing such. The construction of the hierarchical representation in higher dimensions is detailed in [1].

In this paper, we construct a preconditioner, M, for the stiffness matrix from an AFEM discretization of Equation (1.1). This allows us to benefit from the adaptive mesh, without solving the ill-conditioned system. We show that the cost of solving the preconditioned linear system with a Krylov method is competitive with hierarchical GMG approach and furthermore prove that the preconditioned spectrum is clustered around 1, guaranteeing convergence. Our approach also has the advantage of avoiding hierarchical matrices and their sometimes cumbersome implementation.

In the next section, we present background information about structured matrices and preconditioning Toeplitz matrices. Then, in Section 3 we investigate what structure the adaptive mesh retains and how that is reflected in the structure of A. We leverage this structure to apply preconditioning techniques formulated for Toeplitz matrices [5, 6, 4, 8]. This is followed by theoretical results about eigenvalue clustering of the preconditioned system in Section 4. Numerical results to validate our assumptions about the stiffness matrix and compare performance to the GMG method from [19] are presented in Section 5.

2. Background.

2.1. Properties of Structured Matrices. There has long been interest in solving Toeplitz linear systems efficiently. A matrix A is called Toeplitz if $a_{ij} = a_{i-j}$. Arbitrary $n \times n$ matrices have up to n^2 entries and are solved directly in $\mathcal{O}(n^3)$ FLOPS. Since a Toeplitz matrix has at most $2n - 1$ unique entries, we expect to solve Toeplitz systems with $\mathcal{O}(n^2)$ FLOPS, (see for example Levinson's algorithm, Trench's algorithm). Even with these improvements, however, fast Toeplitz algorithms are unstable in general and infeasible for large systems that arise in application [3]. Infinite or periodic Toeplitz matrices can be solved via a transformation to Fourier space and a discrete convolution. On the other hand, finite Toeplitz systems are much more difficult to solve due to the boundaries of the mesh or matrix. To solve these systems, we turn to iterative Krylov and multigrid methods [8, 13, 19]. For these methods we take advantage of Toeplitz structure by using structured preconditioners.

Toeplitz matrices commonly arise in partial differential equations discretization, FDEs, signal processing, and control theory. In application, Toeplitz matrices are often symmetric positive definite (SPD) or symmetric positive semidefinite (SPSD). Given an SPD Toeplitz system $Lx = b$, the idea introduced by Strang in [17] is to use a circulant preconditioner, C, so that $C^{-1}Tx = C^{-1}b$ is solved in fewer iterations. A circulant matrix is a Toeplitz matrix, that additionally has the "wrap-around"

property where the last entry of each row is the first entry of the subsequent row:

$$L = \begin{pmatrix} \ell_0 & \ell_{-1} & \cdots & \ell_{-(n-1)} \\ \ell_1 & \ell_0 & \cdots & \ell_{-(n-2)} \\ \vdots & \vdots & \ddots & \vdots \\ \ell_{n-1} & \ell_{n-2} & \cdots & \ell_0 \end{pmatrix}, \; C = \begin{pmatrix} c_0 & c_1 & \cdots & c_{n-1} \\ c_{n-1} & c_0 & \cdots & c_{n-2} \\ \vdots & \vdots & \ddots & \vdots \\ c_1 & c_2 & \cdots & c_0 \end{pmatrix}. \tag{2.1}$$

As discussed in Section 2.2, the stiffness matrix resulting from AFEM applied to (1.1) is not Toeplitz, but does contain SPD Toeplitz blocks. Toeplitz matrices have many special properties, here we focus on their connection to functions in the Wiener class.

DEFINITION 2.1 (see [5]). *Given a singly infinite, symmetric Toeplitz matrix L we define the finite subsection L_m to be the $m \times m$ principle submatrix. That is,*

$$L = \begin{pmatrix} \ell_0 & \ell_1 & \ell_2 & \\ \ell_1 & \ell_0 & \ell_1 & \ddots \\ \ell_2 & \ell_1 & \ell_0 & \ddots \\ & \ddots & \ddots & \ddots \end{pmatrix}, \; L_m = \begin{pmatrix} \ell_0 & \ell_1 & \cdots & \ell_{m-1} \\ \ell_1 & \ell_0 & \cdots & \ell_{m-2} \\ \vdots & \vdots & \ddots & \vdots \\ \ell_{m-1} & \ell_{m-2} & \cdots & \ell_0 \end{pmatrix}.$$

Assume $\sum_{k=-\infty}^{\infty} |\ell_k| < \infty$, then the matrices induce functions $\tilde{\ell}(z) = \sum_{k=-\infty}^{\infty} \ell_k z^k$ and $\tilde{\ell}_n(z) = \sum_{k=-(n-1)}^{n-1} \ell_k z^k$ are real, positive, and in the Wiener class for $|z| = 1$.

Previous work on Toeplitz systems offers many options for circulant preconditioners, (see [4]). Strang and R. Chan first proposed to construct a circulant C by copying the central diagonals of L into the required wrap-around entries. Assuming for simplicity $n = 2m$, then the first column of C would be $\ell_0, \ldots, \ell_m, \ldots, \ell_0$ [5]. Alternatively, T. Chan constructs a circulant preconditioner, C, by averaging the diagonals of L. He shows this to be optimal in the sense that $||C - L||_F$ is minimal [7]. These circulant preconditioners have become well-known and are sometimes referred to as the G. Strang and T. Chan type circulant preconditioners, respectively.

Although their construction differs, all approaches are diagonalized by some trigonometric transform matrix and are thus efficiently applied via the fast Fourier transform (FFT). The spectrum of the preconditioned systems are asymptotically the same [4]. In this paper, the preconditioner we construct is based on the φ matrix class, introduced in [2].

The φ preconditioner is not itself circulant, but is diagonalized by the type-I discrete sine transform (DST) matrix. Circulant preconditioners, φ, and other preconditioners diagonalized by other discrete trigonometric matrices can all be unified under a displacement operator kernel framework, see [12]. Together these form a class of discrete-trigonometric-based preconditioners. Though our construction could be generalized to circulant preconditioners, we use φ because it does not require complex arithmetic needed for the FFT.

DEFINITION 2.2. *Following [2], given a symmetric $n \times n$ Toeplitz matrix L, we*

build a Hankel correction matrix, H, where

$$L = \begin{pmatrix} \ell_0 & \ell_1 & \ell_2 & \cdots & \ell_{n-2} & \ell_{n-1} \\ \ell_1 & \ell_0 & \ell_1 & \cdots & \ell_{n-3} & \ell_{n-2} \\ \ell_2 & \ell_1 & \ell_0 & \cdots & \ell_{n-4} & \ell_{n-3} \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ \ell_{n-2} & \ell_{n-3} & \ell_{n-4} & \cdots & \ell_0 & \ell_1 \\ \ell_{n-1} & \ell_{n-2} & \ell_{n-3} & \cdots & \ell_1 & \ell_0 \end{pmatrix}$$

$$H = \begin{pmatrix} \ell_2 & \ell_3 & \ell_4 & \cdots & \ell_{n-1} & 0 & 0 \\ \ell_3 & \ell_4 & \ell_5 & \cdots & 0 & 0 & 0 \\ \ell_4 & \ell_5 & \ell_6 & \cdots & 0 & 0 & \ell_{n-1} \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & \ell_5 & \ell_4 & \ell_3 \\ 0 & 0 & \ell_{n-1} & \cdots & \ell_4 & \ell_3 & \ell_2 \end{pmatrix}.$$

Additionally, define $\varphi := L - H$.

Both L and H are symmetric, so they are completely represented by at most n unique values: the diagonals of L : $\ell_0, \ell_1, \ldots, \ell_{n-1}$, and the anti-diagonals of H : $\ell_2, \ell_3, \ldots, \ell_{n-1}, 0, 0$. The following lemma summarizes properties of φ.

LEMMA 2.3 (see [2]). *Let L be a Toeplitz matrix and H the corresponding Hankel correction as in Definition (2.2). Then, $\varphi = L - H$ has the following properties:*

1. *φ is diagonalized by the DST, S. We write $\varphi = S\Lambda S^{-1} = S\Lambda S$.*
2. *φ can be applied in $\mathcal{O}(n \log n)$ FLOPS.*
3. *Let L_1 be the first column of L. Then, the k-th eigenvalue of φ is proportional to the k-th entry of SL_1. Specifically,*

$$\lambda_k(\varphi) = c_k[SL_1]_k, \text{ where } c_k := \sqrt{\frac{n+1}{2}} \frac{1}{\sin(\frac{\pi k}{n+1})}. \tag{2.2}$$

4. *Each eigenvalue of φ is written as the truncated function $\tilde{\ell}_n$ evaluated somewhere on the unit circle, $\lambda_k(\varphi) = \tilde{\ell}_n(z_k)$, where $|z_k| = 1$.*

2.2. AFEM stiffness matrix. Again, let A be the stiffness matrix resulting from AFEM applied to (1.1). During the `MARK` module of AFEM (1.2), it is often the case that adjacent elements have similar error estimates, and are either both marked or both not marked for refinement. As a result, after several refinements there are still subsections of the domain that have been refined to the same level. We refer to these subdomains as *locally uniform*. Since uniform meshes give rise to SPD Toeplitz systems, wherever there are adjacent elements of the same size we find a corresponding Toeplitz block on the main diagonal of the stiffness matrix, A. The size of this block depends on how many adjacent elements are the same size. In practice, we observe the greatest level of refinement near the boundary of the domain. In general, we have larger Toeplitz blocks from locally uniform subdomains in the middle of the matrix, and smaller blocks near the boundary. At the extremes of the locally uniform intervals, we always have interaction between elements of different sizes. This interaction means that Toeplitz blocks are never adjacent along the diagonal, instead there is at least one 1×1 'block' separating them.

These 1×1 blocks, along with their corresponding rows and columns, can be viewed as additional boundaries imposed onto the system. As mentioned earlier, it is

the boundaries of a Toeplitz system that necessitate more computation while solving, so we anticipate that handling of these 1×1 block disruptions to the Toeplitz structure is the key to making a good preconditioner. In order to better handle these 1×1 blocks, we solve the a permuted system, $\tilde{A}$, shifting the rows and columns associated with these blocks to the upper left corner of the matrix, constituting V, W and $W^\top$. That is,

$$PAP^\top = \tilde{A} = \begin{bmatrix} V & W \\ W^\top & Q \end{bmatrix}. \tag{2.3}$$

Consequently, along the diagonal of the lower right block, Q, remain Toeplitz blocks of size at least two. Since A was SPD, the permuted $\tilde{A}$ is also SPD.

Let k denote the size of V, that is the number of 1×1 blocks. It is natural to ask, how big is k relative to $m = \text{size}(A)$? It is difficult to get an exact handle on this theoretically, since it depends on how AFEM routines are implemented. However, in our numerical experiments we find that $k \approx \log(m)$ (see Section 5.1), and k never grows faster than m. With p denoting the number of 1×1 blocks in Q, we might expect that p is exactly one more than k. However, the adaptive refinement carried out in AFEM will occasionally create a few consecutive 1×1 blocks. So while we cannot guarantee an exact relationship between k and p, it is true that $k \approx p$.

3. The Adaptive Preconditioner.

In this section, we set forth the properties we require from a preconditioner. Using φ for its efficient inversion and good approximation of Toeplitz blocks, we construct our preconditioner, M, as an approximation to the LU decomposition of the permuted stiffness matrix.

3.1. Constructing our Preconditioner.

3.1.1. Preconditioner Requirements. A good preconditioner for an iterative method must in general decrease the total number of iterations, without increasing the cost of a single iteration. We adapt criteria for preconditioners from [12], similar criteria is well-established in the literature (for example [2]).

Preconditioner Criteria:

1. Complexity of constructing and applying $M \in \mathbb{R}^{m\times m}$ should be $\mathcal{O}(m \log m)$.
2. A linear system with M should be solved in $\mathcal{O}(m \log m)$ operations.
3. The spectrum of $M^{-1}\tilde{A}$ should be clustered around 1.

3.1.2. LU Decomposition of $\tilde{A}$. To construct our preconditioner we must consider the permuted 1×1 blocks in V and the structured pieces of Q. Using the Schur complement, we obtain a block-LU decomposition for our permuted stiffness matrix. We define the Schur complement matrix $B := Q - W^\top V^{-1} W$, and obtain

$$\tilde{A} = LU = \begin{pmatrix} I & 0 \\ W^\top V^{-1} & I \end{pmatrix} \begin{pmatrix} V & W \\ 0 & B \end{pmatrix}. \tag{3.1}$$

It is straightforward to verify the inverses of the two factor matrices,

$$L^{-1} = \begin{pmatrix} I & 0 \\ -W^\top V^{-1} & I \end{pmatrix},$$

$$U^{-1} = \begin{pmatrix} V^{-1} & -V^{-1} W B^{-1} \\ 0 & B^{-1} \end{pmatrix}.$$

Recalling that $k = \text{size}(V)$ is much smaller and slower-growing than n, inverting V does not contribute to the overall costs. The Sherman-Woodbury-Morris formula allows us to invert B, $B^{-1} = Q^{-1} + Q^{-1}W^\top(V - WQ^{-1}W^\top)^{-1}WQ^{-1}$. This expression, however, requires the prohibitively expensive inversion of Q. Here, we leverage the Toeplitz blocks to build an adequate approximation to Q^{-1}.

3.1.3. Approximating Q^{-1}. Let $L_1, L_2, \ldots, L_p$ be the Toeplitz blocks ordered as they appear along the diagonal of Q^D, with respective sizes $m_1, m_2, \ldots, m_p$. Then for each L_i we construct φ_i as in Definition (2.2). Stacking these blocks we build a block φ matrix as $\Phi := \text{diag}(\varphi_1, \varphi_2, \ldots, \varphi_p)$. By repeated use of Lemma 2.3 on each φ_i, we apply Φ^{-1} implicitly. Let $m := \max_j m_j$, then the cost of applying Φ^{-1} is bounded by

$$\begin{aligned}\mathcal{O}\left(\sum_{j=1}^{k} m_j \log(m_j)\right) &\leq \mathcal{O}\left(m \sum_{j=1}^{k} \log(m)\right) \\ &\leq \mathcal{O}\left(km \log(m)\right).\end{aligned}$$

Let S_{m_i} denote the type-I DST matrix of size m_i. Using Equation (2.2) to calculate the eigenvalues of φ_i, we write $\varphi_i = S_{m_i}\Lambda_i S_{m_i}$. Then,

(3.2)
$$\Phi^{-1} = \begin{pmatrix} S_{m_1} & & & \\ & S_{m_2} & & \\ & & \ddots & \\ & & & S_{m_p} \end{pmatrix} \begin{pmatrix} \Lambda_1^{-1} & & & \\ & \Lambda_2^{-1} & & \\ & & \ddots & \\ & & & \Lambda_p^{-1} \end{pmatrix} \begin{pmatrix} S_{m_1} & & & \\ & S_{m_2} & & \\ & & \ddots & \\ & & & S_{m_p} \end{pmatrix}.$$

The DST blocks in the outside matrices are be applied via the fast sine transform (FST) to do matrix multiplication in $\mathcal{O}(m \log m)$ FLOPS. The eigenvalue matrix is diagonal and inverted in m operations, and can be multiplied against a Toeplitz matrix embedded in a $2m \times 2m$ circulant matrix in $\mathcal{O}(m \log m)$ FLOPS. Moreover, both the FST and block arithmetic are highly parallelizable for more efficient implementation. This establishes Criteria (1) and (2) for Φ. Therefore Φ^{-1} is fast to compute, and in Section 4 we show it is a good spectral approximation to Q^{-1}.

3.1.4. Defining the Adaptive Preconditioner, M^{-1}. Let $\hat{B} = \Phi - W^\top V^{-1} W$. Then, define

$$\hat{U} := \begin{pmatrix} V & W \\ 0 & \hat{B} \end{pmatrix}, \; \hat{U}^{-1} = \begin{pmatrix} V^{-1} & -V^{-1}W\hat{B}^{-1} \\ 0 & \hat{B}^{-1} \end{pmatrix}.$$

Our preconditioner is given by

$$\begin{aligned} M^{-1} := \hat{U}^{-1}L^{-1} &= \begin{pmatrix} V^{-1} & -V^{-1}W\hat{B}^{-1} \\ 0 & \hat{B}^{-1} \end{pmatrix} \begin{pmatrix} I & 0 \\ -W^\top V^{-1} & I \end{pmatrix} \\ &= \begin{pmatrix} V^{-1} + V^{-1}W\hat{B}^{-1}W^\top V^{-1} & -V^{-1}W\hat{B}^{-1} \\ -\hat{B}^{-1}W^\top V^{-1} & \hat{B}^{-1} \end{pmatrix}. \end{aligned}$$

The cost of applying M^{-1} is dominated by application of the biggest block, $\hat{B}^{-1}$. However, by using Φ in place of Q, and consequently $\hat{B}$ in place of B, we use the Sherman-Morrison-Woodbury formula to calculate $\hat{B}^{-1} = (\Phi - \tilde{W}^\top \tilde{V}^{-1}\tilde{W})^{-1} = (\Phi^{-1} + \Phi^{-1}W^\top(V - W\Phi^{-1}W^\top)^{-1}W\Phi^{-1}$. Thus, using that fast application of Φ, as explained above, M^{-1} can be applied in $\mathcal{O}(m \log m)$ FLOPS. Therefore M also satisfies Criteria (1) and (2).

4. Theoretical Results. How tightly the eigenvalues cluster around 1 determines the speed of convergence. In this section we prove that the spectral clustering in Criterion (3) holds for M. Proofs for clustering of discrete-trigonometric-transform based preconditioned problems in this area tend to follow a similar structure: show the preconditioned matrix can be written as the sum of the identity matrix, a small-norm matrix, and a low-rank matrix. The small-norm term has spectrum clustered around 0, forcing the clustering of the entire system around 1, with the number of outliers bounded by the rank of the low-rank term. We build up to the clustering of the final preconditioned system in several steps. First, we briefly present our version of the spectral clustering proof for the case of φ acting on a single Toeplitz block. Then we expand this result to prove the clustering of $\Phi^{-1}Q$; that is the block Φ matrix applied to the Toeplitz blocks and corresponding off diagonal blocks of Q. Finally, we show how our handling of the 1×1 blocks preserves the spectral clustering of the whole system, $M^{-1}\tilde{A}$. The following lemma will be of use to us throughout the section.

LEMMA 4.1. *Let $T \in \mathbb{R}^{n\times n}$ be symmetric positive definite. Let $A \in \mathbb{R}^{n\times n}$ also be symmetric positive definite, then*

$$\lambda_k(T^{-1}A) \leq \frac{\lambda_k(A)}{\lambda_{\min}(T)}.$$

Proof. Recalling that $\lambda_k(T^{-1}A) = \lambda_k(AT^{-1})$, and that since both matrices are SPD the singular values and eigenvalues coincide. Then we apply Theorem 3.3.16 (d) from [10],

$$\lambda_k(AT^{-1}) \leq \lambda_k(A)\lambda_{\max}(T) = \frac{\lambda_k(A)}{\lambda_{\min}(T)}.$$

4.1. Spectral Clustering of a Toeplitz Block. Proofs demonstrating the eigenvalue clustering of a circulant or trigonometric-based preconditioners on a Toeplitz matrices can follow a few strategies, here we briefly give a version, which utilizes the infinite matrix framework set forth in Section 2.1. Proofs in [2] and [12] follow a similar technique, see [5] for an approach relying on the infinite matrix framework and theory of collectively compact operators. In keeping with these proofs, we assume the generating function $\tilde{\ell}^{(i)}(z)$ induced by each Toeplitz block L_i is in the Wiener class and positive on the unit circle. That is, for all $1 \leq i \leq p, \tilde{\ell}^{(i)}(z) = \sum_{j=-\infty}^{\infty} |\ell_j| < \infty$ and $\ell^{(i)}(z) > 0$ for $|z| = 1$.

4.1.1. Approximating Q^{-1}.

LEMMA 4.2 (Uniform positivity on the unit circle). *Let L be a singly infinite, symmetric, Toeplitz matrix with diagonals $\ell_0, \ell_1, \ldots$, and L_m is its $m \times m$ truncation. Assume the generating function $\tilde{\ell}(z)$ is in the Wiener class and positive on the unit circle. Then, for $|z| = 1$, $\tilde{\ell}_m(z) > \varepsilon$.*

Proof. Since $\tilde{\ell}(z) > 0$ for $|z| = 1$, compactness of the unit circle implies that for some $\varepsilon > 0$, $\tilde{\ell}(z) > 2\varepsilon$ for $|z| = 1$. By the Wiener class assumption, choose N such that $\sum_{j=N}^{\infty} \ell_j z^j \leq \sum_{j=N}^{\infty} |\ell_j| < \varepsilon$. So for all $m \geq N$,

$$2\varepsilon < \ell(z) = \tilde{\ell}_m(z) + \sum_{j=m}^{\infty} \ell_j z^j < \tilde{\ell}_m(z) + \varepsilon.$$

LEMMA 4.3 (Eigenvalue Clustering on a Toeplitz Block, [2, 12]). *Let L, L_m, $\tilde{\ell}$, and $\tilde{\ell}_m$ be as above, and φ as in Definition 2.2. Then for m large enough, the spectrum of $\varphi^{-1}L$ is clustered around 1.*

Proof. Since $\varphi = L - H$, $\varphi^{-1}L = \varphi^{-1}(\varphi + H) = I + \varphi^{-1}H$, it suffices to show that the eigenvalues of $\varphi^{-1}H$ are clustered around 0.
We split the Hankel matrix into a low-rank and small-norm piece, $H = H^{LR} + H^{SN}$. Where H^{LR} contains the anti-diagonals $\ell_2, \ldots, \ell_N$ with $s := \text{rank}(H^{LR}) \ll m$. The small-norm descriptor is justified for $H^{SN} = H - H^{LR}$, a Hermitian $m \times m$ matrix with at most two copies of $\ell_N, \ldots, \ell_{m-1}$ in each row and column. Thus $||H^{SN}||_2 = \sqrt{||H^{SN}||_1 ||H^{SN}||_\infty} = ||H^{SN}||_1 < 2\varepsilon$. Hence by Weyl's Inequality, the eigenvalues of H are clustered within 2ε of zero with at most s outliers.
Now by Lemma 4.1,

$$\lambda_k(\varphi^{-1}H) \leq \lambda_k(H)\frac{1}{\lambda_{\min}(\varphi)} = \lambda_k(H)\frac{1}{\tilde{\ell}_m(z_{\min})} \leq \lambda_k(H)\frac{1}{\varepsilon}. \tag{4.1}$$

Thus, the clustering of the spectrum of H around 0 implies the same for $\varphi^{-1}H$. ∎

Equation (4.1) shows that the bigger $\lambda_{\min}(\varphi)$, the stronger the bound. We calculate this eigenvalue exactly using Lemma 2.3, and Equation (2.2). It can also be bounded in terms independent of φ, using the generating function $\tilde{\ell}(z)$, as above.

4.2. Spectral Clustering of the Block System. We now prove the spectral clustering of $\Phi^{-1}Q$.

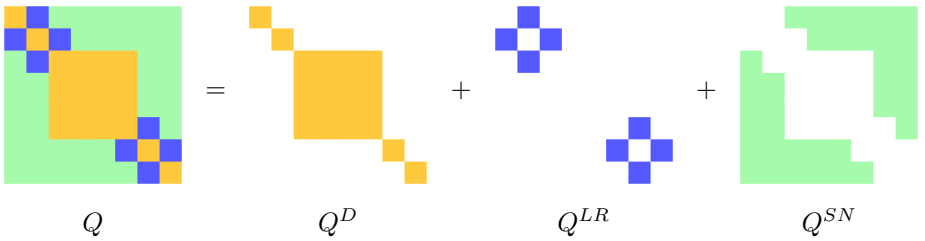


Fig. 1: Illustration of the splitting of Q: Q^D is block diagonal with Toeplitz blocks, blocks in Q^{LR} are small and together form a low-rank matrix, and Q^{SN} is assumed to be small-norm due to weak interaction between elements.

LEMMA 4.4. *Let A be the stiffness matrix as described in Section 2.2 and Φ as described in Section 3. Then, the spectrum of $\Phi^{-1}Q$ is clustered around 1.*

Proof. It is easy to generalize the lemmas above to the block diagonal Φ^{-1} acting only on the Toeplitz blocks, Q^D. The work in this proof is in showing that $\Phi^{-1}(Q^{LR} + Q^{SN})$ can also be split into small-norm and low-rank pieces. To begin, we further breakdown Q into three summands, $Q = Q^D + Q^{LR} + Q^{SN}$, where Q^D contains the Toeplitz blocks, Q^{LR} is low-rank ,and Q^{SN} is small-norm.
Due to the permutation, Q has Toeplitz blocks of size at least 2 along its diagonal. We use Q^D to denote the block diagonal with Toeplitz blocks.
The fractional derivative results in a dense matrix, but we observe off-diagonal decay, the strength of which depends on the fractional index α. Elements sufficiently far from the diagonal are physically separated and have weak interaction. However,

merely excluding the Toeplitz blocks is not enough to get a small-norm matrix. As visualized in Figure 1, outside of the Toeplitz blocks, the entries closest to the diagonal occur at the boundary between Toeplitz blocks. We form small blocks from these adjacent-to-the-diagonal entries, comprising our low-rank term, Q^{LR}. The rank of which is bounded by the size of the block we take, s, times the number of blocks, which is the same as the number of 1×1 blocks, k. That is, $\text{rank}(Q^{LR}) \leq sk$. Finally, the remaining entries makeup our small-norm matrix Q^{SN}. We might naturally ask how big s should be? The rank of Q^{LR} contributes to the number of eigenvalue outliers, while the norm of Q^{SN} determines the strength of the clustering around 1. So a bigger s would indicate more outliers but a tighter cluster, and a smaller s indicates less outliers but a broader cluster. This is discussed more in Section 5, but numerical tests indicate a good estimate for s falls around 4.
Now we show the eigenvalues of $\Phi^{-1}Q = \Phi^{-1}(Q^D + Q^{LR} + Q^{SN})$ are clustered around 1. For each Toeplitz block L_i in $Q^D = \text{diag}(L_1, \dots L_p)$, we use definition 2.2 to construct a Hankel correction matrix, H_i. Let $H = \text{diag}(H_1, \dots H_p)$. Then we can extend the final part of definition 2.2 to its block diagonal counterpart, $\text{diag}(\varphi_1, \dots \varphi_p) = \text{diag}(L_1, \dots L_p) - \text{diag}(H_1, \dots H_p)$ or $Q^D = \Phi + H$. Thus, $Q = \Phi + H + Q^{LR} + Q^{SN}$ and

$$\Phi^{-1}Q = \Phi^{-1}(\Phi + H + Q^{LR} + Q^{SN}) = I + \Phi^{-1}(H + Q^{LR} + Q^{SN}). \tag{4.2}$$

Splitting H is done via application of the splitting in the proof of Lemma 4.4 to each block. As before, we have $\varepsilon > 0$ and choose N_i smaller than each m_i so that for every Toeplitz block the truncated function satisfies $a_{m_i}(z) > \varepsilon$ on the unit circle and $\sum_{i=N_i}^{\infty} |a_i| < \varepsilon$. Then $H = H^{LR} + H^{SN}$, by separating into H^{LR} the anti-diagonals from block i with coefficients $a_0, \dots, a_{N_i}$ and into H^{SN} the anti-diagonals comprising of $a_{N_i+1}, \dots a_{m_i}$ for each block. We note two important bounds: $||H^{SN}||_2 < \varepsilon$ and $\text{rank}(H^{LR}) \leq 2p(N-1)$, where p is the number of Toeplitz blocks and $N := \max_i N_i$. Equation (4.2) is now be written as

$$\Phi^{-1}Q = I + \Phi^{-1}(H^{LR} + Q^{LR}) + \Phi^{-1}(H^{SN} + Q^{SN}). \tag{4.3}$$

This is ostensibly the low-rank plus small-norm representation we sought, but we finish the proof by giving explicit bounds on the rank and norm of these terms. Since Φ is full-rank, $\text{rank}(\Phi^{-1}(H^{LR} + Q^{LR})) = \text{rank}(H^{LR} + Q^{LR})$. Now using each term's individual bounds,

$$\text{rank}(H^{LR} + Q^{LR}) \leq \text{rank}(H^{LR}) + \text{rank}()Q^{LR}) \leq 2p(N-1) + sk,$$

and it remains to show $\Phi^{-1}(H^{SN} + Q^{SN})$ is small norm.

$$||\Phi^{-1}(H^{SN} + Q^{SN})||_2 \leq ||\Phi^{-1}H^{SN}||_2 + ||\Phi^{-1}Q^{SN}||_2.$$

The block multiplication $\Phi^{-1}H^{SN}$ is each φ acting on its corresponding Hankel correction. Since the matrices are SPD,
$||\Phi^{-1}H^{SN}||_2 = \lambda_{\max}(\Phi^{-1}H^{SN}) \leq \frac{1}{\varepsilon}\lambda_{\max}(H)$, by Lemma 4.3. Next,

$$||\Phi^{-1}Q^{SN}||_2 \leq ||\Phi^{-1}||_2||Q^{SN}||_2 = \sigma_{\max}(\Phi^{-1})\sigma_{\max}(Q^{SN}) = \frac{\sigma_{\max}(Q^{SN})}{\lambda_{\min}(\Phi)}.$$

Since Q^{SN} is small-norm this value is small,

$$||\Phi^{-1}(H^{SN}+Q^{SN})||_2 \leq ||\Phi^{-1}H^{SN}||_2 + ||\Phi^{-1}Q^{SN}||_2$$
$$\leq \frac{1}{\varepsilon}\lambda_{\max}(H) + \frac{\sigma_{\max}(Q^{SN})}{\lambda_{\min}(\Phi)}. \qquad \square$$

The strength of this bound depends on $\lambda_{\min}(\Phi)$, but can be written independently of Φ in terms of the generating functions as in Equation (4.1). This dependence is the same for the single block case, and appears in similar proofs for discrete-trigonometric preconditioners (see for example [12]).
In practice the block size, m_i, is determined by the AFEM algorithm, which in turn dictates the actual size ε and hence the strength of clustering. Thus, bigger Toeplitz blocks in A result in tighter eigenvalue clustering.

REMARK 1. *Note that extending the proof from one Toeplitz block to the adaptive stiffness matrix relies only on being able to split the block into a small-norm and a low-rank piece. Thus, nothing about this proof is overly-specific to the φ preconditioner and the methods could easily be extended to other types of circulant preconditioners given in [5, 6].*

4.3. Spectral Clustering of the Preconditioned System. To complete this final proof we require the additional assumption that $||(\Phi^{-1}W^\top(V-W\Phi^{-1}W^\top)^{-1}W)\Phi^{-1}||_2$ is bounded. See Table 1 for numerical validation of this assumption.

THEOREM 4.5. *Following the established notation, assume $\Phi^{-1}(H^{SN}+Q^{SN}) < \varepsilon$ and define $s := rank(\Phi^{-1}(H^{LR}+Q^{LR}))$. Assume also that $||(\Phi^{-1}W^\top(V-W\Phi^{-1}W^\top)^{-1}W)\Phi^{-1}||_2$ is bounded, then the spectrum of $M^{-1}A$ is clustered around 1, with at most s outliers.*

Proof. Let B and $\hat{B}$ be defined as in Section 3.1.4. By matrix multiplication and some simplification we get that the preconditioned system has the form:

$$M^{-1}\tilde{A} = \begin{pmatrix} V^{-1}+V^{-1}W\hat{B}^{-1}W^\top V^{-1} & -V^{-1}W\hat{B}^{-1} \\ -\hat{B}^{-1}W^\top V^{-1} & \hat{B}^{-1} \end{pmatrix}\begin{pmatrix} V & W \\ W^\top & Q \end{pmatrix}$$
$$= \begin{pmatrix} I & V^{-1}W\hat{B}^{-1}(\Phi-Q) \\ 0 & \hat{B}^{-1}(Q-W^\top V^{-1}W) \end{pmatrix}.$$

Since this is an upper-triangular system with a $k\times k$ identity block on the diagonal, we know that 1 is an eigenvalue of the preconditioned system, with algebraic multiplicity at least k, and the remaining $n-k$ eigenvalues are the spectrum of $\hat{B}^{-1}(Q-W^\top V^{-1}W)$. By the Sherman-Morrison-Woodbury formula $\hat{B}^{-1} = (\Phi - \tilde{W}^\top\tilde{V}^{-1}\tilde{W})^{-1} = (\Phi^{-1}+\Phi^{-1}W^\top(V-W\Phi^{-1}W^\top)^{-1}W\Phi^{-1}$, which we applied against $Q-W^\top V^{-1}W$ is

$$\hat{B}^{-1}(Q-W^\top V^{-1}W) =$$
$$\Phi^{-1}Q + \Phi^{-1}W^\top(V-W\Phi^{-1}W^\top)^{-1}W\Phi^{-1}Q$$
$$-\Phi^{-1}W^\top V^{-1}W - \Phi^{-1}W^\top(V-W\Phi^{-1}W^\top)^{-1}W\Phi^{-1}W^\top V^{-1}W.$$

We previously showed $\Phi^{-1}Q = I + \Phi^{-1}(H^{SN} + Q^{SN}) + \Phi^{-1}(H^{LR} + Q^{LR})$, so now

$$\begin{aligned}
&\hat{B}^{-1}(Q - W^\top V^{-1} W) = \\
&\quad I + \Phi^{-1}(H^{SN} + Q^{SN}) + \Phi^{-1}(H^{LR} + Q^{LR}) \\
&\quad + \big[\Phi^{-1}W^\top (V - W\Phi^{-1}W^\top)^{-1} W (I + \Phi^{-1}(H^{SN} + Q^{SN}) \\
&\quad + \Phi^{-1}(H^{LR} + Q^{LR})) \\
&\quad - \Phi^{-1}W^\top V^{-1} W - \Phi^{-1}W^\top (V - W\Phi^{-1}W^\top)^{-1} W \Phi^{-1} W^\top V^{-1} W\big].
\end{aligned}$$

We set aside the terms

$$I + \Phi^{-1}(H^{SN} + Q^{SN}) + \Phi^{-1}(H^{LR} + Q^{LR}) \tag{4.4}$$

$$+ (\Phi^{-1}W^\top (V - W\Phi^{-1}W^\top)^{-1} W)(\Phi^{-1}(H^{SN} + Q^{SN})) \tag{4.5}$$

$$+ (\Phi^{-1}W^\top (V - W\Phi^{-1}W^\top)^{-1} W)(\Phi^{-1}(H^{LR} + Q^{LR})), \tag{4.6}$$

and show the remaining terms cancel out:

$$\begin{aligned}
&\Phi^{-1}W^\top (V - W\Phi^{-1}W^\top)^{-1} W - \Phi^{-1}W^\top V^{-1} W \\
&- \Phi^{-1}W^\top (V - W\Phi^{-1}W^\top)^{-1} W \Phi^{-1} W^\top V^{-1} W \\
&= [-\Phi^{-1}W^\top V^{-1} + \Phi^{-1}W^\top (V - W\Phi^{-1}W^\top)^{-1}(I - W\Phi^{-1}W^\top V^{-1})] W \\
&= [-\Phi^{-1}W^\top V^{-1} + \Phi^{-1}W^\top (V - W\Phi^{-1}W^\top)^{-1}(V - W\Phi^{-1}W^\top) V^{-1}] W \\
&= [-\Phi^{-1}W^\top V^{-1} + \Phi^{-1}W^\top V^{-1}] W = 0.
\end{aligned}$$

The clustering depends on the spectrum of the terms in Equations (4.4) - (4.6). The non-identity terms in Equation (4.4) are small-norm and low-rank from Lemma 4.4. Examining Equation (4.5), by assumption $||(\Phi^{-1}W^\top (V - W\Phi^{-1}W^\top)^{-1} W)\Phi^{-1}||_2$ is bounded; our previous results showed that $H^{SN} + Q^{SN}$ is small-norm. Thus $(\Phi^{-1}W^\top (V - W\Phi^{-1}W^\top)^{-1} W)\Phi^{-1}(H^{SN} + Q^{SN})$ is small-norm. Finally, since we know $\text{rank}(\Phi^{-1}(H^{LR} + Q^{LR})) \leq s$, Equation (4.6) also has rank at most s. Therefore the spectrum of the preconditioned system is the same as the spectrum of the matrix in Equations (4.4) - (4.6), which we have shown to be decomposable as the identity matrix plus a small-norm matrix and a matrix with rank bounded by s.□

The spectral clustering described in Criterion 3 is therefore satisfied by $M^{-1}\tilde{A}$. Since Φ is diagonalized by the DST matrix, it can be constructed and Φ^{-1} can be applied in $\mathcal{O}(m \log m)$. In Section 3.1.4 we showed that this speed-up extends to constructing and applying M^{-1}, satisfying Criteria 1 and 2 for M. We have now shown that M satisfies all three criteria we sought in a preconditioner.

5. Numerical Results. To validate the theoretical results and test the performance of our preconditioner, we ran tests to solve Equation (1.1) with different right hand sides. We used preconditioned GMRES and solved to a tolerance of $\varepsilon = 10^{-7}$. In each case, we tested fractional indices $\alpha = 1.1, 1.2, \ldots, 1.9$, paying special attention to $\alpha = 1.9$ since the condition number of the problem worsens as α approaches two.
All code was written in MATLAB and run on a MacBook Pro laptop with M1 Pro chip and 16 GB RAM.

5.1. Validating Assumptions.

In building our preconditioner, we claimed that inverting V is relatively cheap since its size k, the number of 1×1 blocks , is smaller and grows more slowly than m. In pursuit of validating this assumption we must consider the mechanism by which AFEM refines the mesh. If AFEM is run to a set level of refinement, sufficient accuracy may be attained several levels before termination. Successive refinements will inflate k, with little return in improved accuracy. Thus to best measure how k grows relative to m, we solved Equation (1.1) with a known solution, $u(x) = 10x^2(1-x^2)$ (see [19] example 6.1 for more details) across values of α, each time refining only until the L^2 error was below $\varepsilon = 10^{-7}$.

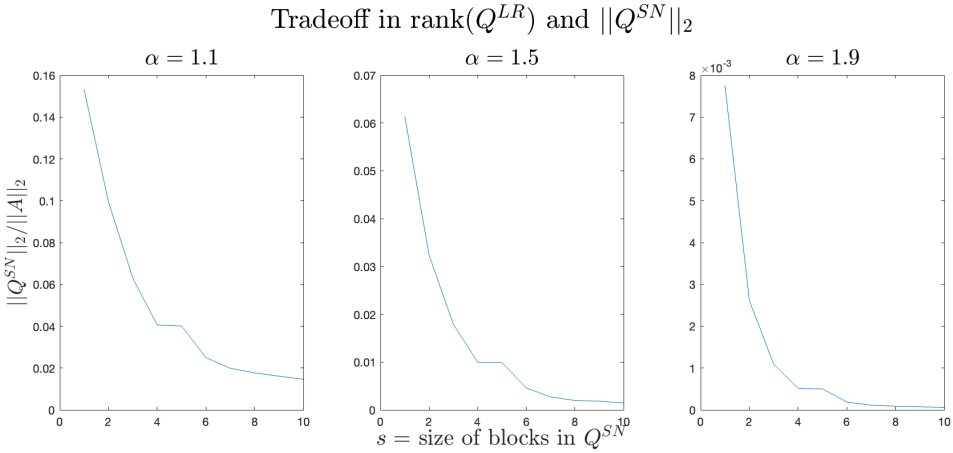


Fig. 2: The relative norm of Q^{SN} is plotted against s, the size of blocks in Q^{LR}. Initially, increasing s drastically lowers $||Q^{SN}||_2$, but then levels off.

In order to prove the spectral clustering, we made two assumptions about the structure of Q: the blocks adjacent to the diagonal are well-approximated by low-rank matrices and the other off-diagonal blocks are small-norm. The size of the blocks in Q^{LR} has an inverse relationship to $||Q^{SN}||_2$. The results in Figure 2 visualize this relationship for solving Equation (1.1) to a tolerance of $\varepsilon = 10^{-7}$, across various fractional indices, α. The exact solution on this example is $u(x) = 10x^2(1-x^2)$ (see [19] example 6.1 for more details). Recall that rank(Q^{LR}) bounds the number of outliers and $||Q^{SN}||_2$ relates to a bound on the clustering radius. We interpret a larger s as classifying more eigenvalues as outliers and a very tight cluster around 1, whereas a smaller s and consequently larger $||Q^{SN}||_2$ would mean more eigenvalues are considered part of the cluster, but the cluster itself is bigger. This choice is a matter of interpretation for the theoretical proofs and not necessary for using the preconditioner, so it suffices to demonstrate that s and $||Q^{SN}||_2$ are simultaneously sufficiently small, as seen in Figure 2.

update table with fixed numbers

In the proof of Theorem 4.5, we needed $(\Phi^{-1}W^\top(V - W\Phi^{-1}W^\top)^{-1}W)\Phi^{-1}(H^{SN} + Q^{SN})$ to be small-norm, which depended on the assumption that $||(\Phi^{-1}W^\top(V - W\Phi^{-1}W^\top)^{-1}W)\Phi^{-1}||_2$ was bounded. Since $||H^{SN} + Q^{SN}||_2$ is small (though its exact value depends on how ε is chosen in the proof of Lemma 4.4), we need only demonstrate that the other part of

Table 1: $||(\Phi^{-1}W^\top(V - W\Phi^{-1}W^\top)^{-1}W)\Phi^{-1}||_2$

Refinement level	2	3	4	5	6	7	8	9
$\alpha = 1.1$	1.42	5.85	3.86	64.6	32.137	12.406	28.14	N/A
$\alpha = 1.3$	0.55	9.45	5.19	2.45	24.142	118.86	76.59	10.304
$\alpha = 1.5$	0.77	0.91	1.57	10.085	23.442	1.19	1.270	16.43
$\alpha = 1.7$	0.19	0.50	0.12	0.419	0.28575	0.431	137.48	873.738
$\alpha = 1.9$	0.01	0.03	0.007	0.0092	0.0101	1.07	0.05	132.31

$||(\Phi^{-1}W^\top(V - W\Phi^{-1}W^\top)^{-1}W)\Phi^{-1}||_2$ for various α, up to the refinement necessary for a solution with L^2 error below $\varepsilon = 10^{-7}$. Refinement level 1 is not included since it represents a uniform grid, where our preconditioner reduces to the Bini and de Benedetto preconditioner.

this term, $||(\Phi^{-1}W^\top(V - W\Phi^{-1}W^\top)^{-1}W)\Phi^{-1}||_2$ doesn't blow-up. Example results of this value are given in Table 1. Though the norm of this term can vary somewhat, it never jumps orders of magnitude, and is often below 1. This also supports a tight eigenvalue cluster, since this term is part of the small-norm matrix that determines the clustering radius.

5.2. Performance Results. Example results of the spectrum before and after preconditioning are in Figure 3. These tests are for the problem (1.1) with $\alpha = 1.9$ and exact solution $u(x) = 10x^2(1 - x^2)$, as before. Note that although the proof allows for eigenvalue outliers far beyond the cluster, in practice everything remains within the interval $(0, 2)$. Even for the most ill-conditioned values of α, the preconditioner vastly reduces the spectral radius of the system and further produces a cluster around 1. After we exclude the p eigenvalues farthest from 1, we get a very tight cluster, in this example between 0.94 and 1.03. This suggests that the system should be near convergence after p iterations. Indeed, Figure 4 shows that the number of iterations required to converge closely follows the number of Toeplitz blocks in our stiffness matrix. Their approximate equality further supports the interpretation of 1×1 blocks acting as additional boundaries imposed on the problem, requiring roughly one iteration for each outlier eigenvalue these boundaries cause.

Finally, we compare our preconditioner to the hierarchical GMG method in [19]. Figure 5 shows that our estimated error closely follows that of the GMG method. Thus using M with preconditioned GMRES is as accurate and as fast as hierarchical methods for the adaptive problem. Our approach has the advantage of provable eigenvalue bounds for the preconditioned system and a straightforward implementation that does not require building a hierarchical representation of the stiffness matrix nor use of hierarchical matrix operations. In turn, the close match of L^2 error in Figure 5 suggests that both methods converge to the same solution, and offers further evidence that hierarchical GMG is robust.

6. Conclusion. The singularities inherent to FDEs require adaptive meshes to approximately distribute error across intervals. The stiffness matrix resulting from

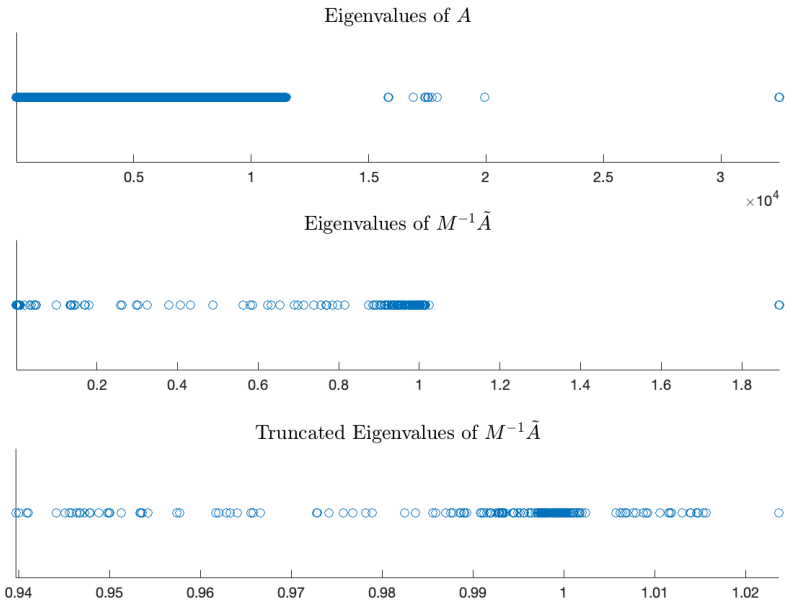


Fig. 3: Eigenvalue clustering before and after preconditioning for $m = 4417$ and $\alpha = 1.9$, note the scale on the top plot. The bottom plot truncates the p eigenvalues furthest from one, suggesting the number of Toeplitz blocks bounds the number of outliers in practice.

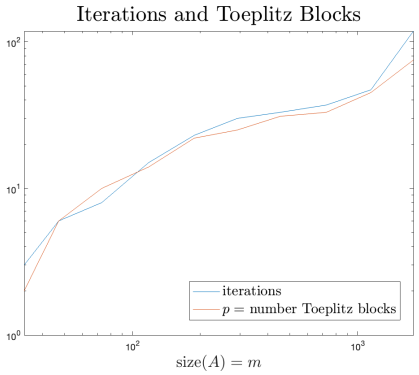
AFEM is ill-conditioned. Previous work allows for hierarchical representation of A, enabling faster computation for solving with GMG [19, 1]. We have offered an alternative approach, leveraging remaining structure in A and utilizing well-known preconditioners for Toeplitz matrices. We have shown our preconditioner to be efficient to build, apply, and effective at reducing the number of Krylov iterations needed to solve the linear system. In numerical tests, we perform competitively against GMG and additionally have theoretic proof of accelerated convergence. Future work would extend these ideas to FDEs in higher dimensions. We expect the preconditioner would perform well in this setting, though theoretical results are much harder to obtain.

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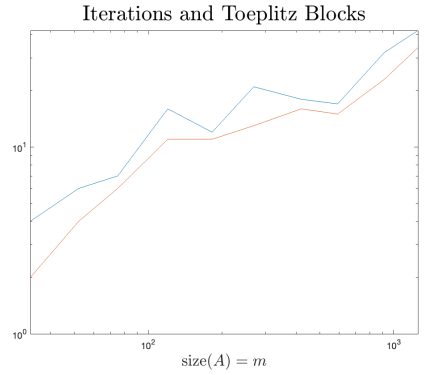
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(a) Solving Equation (1.1), where the exact solution is $u(x) = 10x^2(1 - x^2)$.



(b) Solving Equation (1.1), with $f(x) = -(1 + \sin(x))$.

Fig. 4: Iterations to convergence below a tolerance of $\varepsilon = 10^{-7}$ for $\alpha = 1.9$, plotted with the number of Toeplitz blocks across increasing matrix size m.

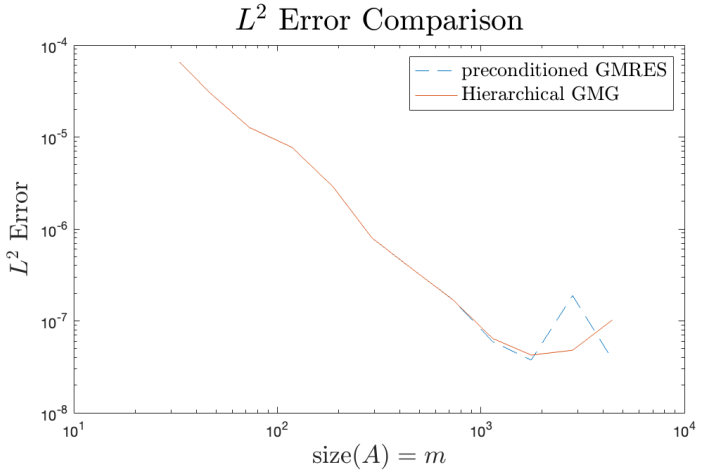


Fig. 5: L^2 error comparison with Hierachical GMG method for $\alpha = 1.9$ and where the exact solution is $u(x) = 10x^2(1 - x^2)$.

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